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Online parameter identification of permanent magnet synchronous motor based on recursive least squares algorithm

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Abstract. In order to solve the problem of variable parameters in the system when the PMSM works, this project intends to use the recursive algorithm in the steady state of alpha-beta. This project identifies the two parameters as the fixed values of two parameters at the same time, so as to reduce the number of parameters to be estimated and avoid the problem of under-classification. The proposed method is studied experimentally on the 2 kW PMSM experimental platform. Simulation results show that this method can quickly and accurately realize the online identification of parameters in the PMSM system.

1. Introduction

A vector control system for the PMSM often relies on relatively accurate motor parameters, such as motor inductance, resistance, and magnetic chain.

Offline identification is the use of experimental means to obtain the parameters of the motor, generally using a certain excitation method, or letting the motor run under certain conditions [1], and then according to the response of the parameter calculation. However, in offline parameter identification, it is often necessary to obtain more signals with other devices, which leads to more cumbersome data collection and storage in the system.

In view of the shortcomings of traditional offline recognition technology, online recognition technology is widely used at present. For the online parameter identification method, the results not only accurately reflect the change of motor parameters with the operating state of the system, but also can be applied to the adaptive control. Online methods such as recursive least squares [2], Kalman filtering [3], model reference adaptation [4], and radiative projection algorithms [5] are often used to update parameters in real-time. In [6], a parameter identification algorithm based on the recursive least square method is proposed, which has the advantages of a simple algorithm, a small amount of calculation, and fast convergence.

In this paper, the least square method is used to identify the PMSM in real time. Under static coordinates, a mathematical model established in the PMSM is established, and its stator resistance, dq axis inductance, and magnetic flux are identified, using the observable electrical quantities such as voltage and current of the motor as input quantities. At the same time, the under-ranking problem of the parameter identification of the PMSM is considered. There are only two PMSM voltage equations but four motor parameters in the equations. The four parameters cannot be identified at the same time in the steady state of the motor, so two of the parameters are used as fixed values to identify the other two parameters in this paper to avoid the mis-convergence or non-convergence of the identification

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results caused by adopting the under-ranking identification model for full parameter identification. The problem is avoided.

2. Online parameter identification strategy

2.1. Recursive least squares algorithm

The least squares method is chosen as the algorithm which is an algorithm to find the best matching value of the unknowns by minimizing the sum of squares of errors. Its linear system can be described as follows:

$$y(k) = \theta^T(k)\Phi(k) + \xi(k) \tag{1}$$

where *y(k)* is the system output at moment k; $\theta^T(k) = [\theta_1(k), \theta_2(k), ..., \theta_n(k)]$, the unknowns to be identified; $\Phi(k) = [\Phi_1(k), \Phi_2(k), ..., \Phi_n(k)]^T$ is the system input at moment k; And $\xi(k)$ is the model residual.

The error sum of squares is defined as a criterion function:

$$J = \sum_{k=1}^{k}\left[y(k) - \theta^T(k)\Phi(k)\right]^2 \tag{2}$$

Using multiple sets of input and output data, the value *J* of the criterion function is minimized, which is the estimation result of the algorithm.

The formula for the recursive least squares method is as follows:

$$\begin{aligned} &\hat{\theta}(k) = \hat{\theta}(k-1) + K(k)[y(k) - \hat{\theta}^T(k-1)\Phi(k)] \\ &K(k) = P(k-1)\Phi(k)[\lambda + \Phi^T(k)P(k-1)\Phi(k)]^{-1} \\ &P(k) = [I - K(k)\Phi^T(k)]P(k-1)\lambda^{-1} \end{aligned} \tag{3}$$

Under the alpha-beta static coordinate system, the mathematical modeling of the PMSM can be obtained from Equation (1) through Clark transformation:

$$\begin{aligned} \begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} = \begin{bmatrix} R_s + \rho L_d & \omega_e(L_d - L_q) \\ -\omega_e(L_d - L_q) & R_s + \rho L_q \end{bmatrix} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \\ [(L_d - L_q)(\omega_e - \rho i_q) + \omega_e \psi_f] \begin{bmatrix} -\sin\theta_e \\ \cos\theta_e \end{bmatrix} \end{aligned} \tag{4}$$

From Equation (1), it can be seen that the use of the least squares method requires the determination of three matrices of output, input, and the quantity to be identified:

$$\begin{aligned} u_\alpha - \hat{\theta}_{\alpha 2}\Phi_{\alpha 2} = \hat{\theta}_{\alpha 1}\Phi_{\alpha 1} \\ u_\beta - \hat{\theta}_{\beta 2}\Phi_{\beta 2} = \hat{\theta}_{\beta 1}\Phi_{\beta 1} \end{aligned} \tag{5}$$

where

$$\begin{aligned} \hat{\theta}_{\alpha 1} = \hat{\theta}_{\beta 2} = [\hat{R}_s\ \hat{L}_d] \\ \hat{\theta}_{\alpha 2} = \hat{\theta}_{\beta 1} = [\hat{L}_q\ \hat{\psi}_f] \end{aligned} \tag{6}$$

$$
\begin{aligned}
\Phi_{\alpha1} &= \begin{bmatrix} i_\alpha \\ \frac{1}{2}\rho i_\alpha + (\frac{1}{2}\rho i_\beta - \omega_e i_\alpha)sin2\theta_e + (\frac{1}{2}\rho i_\alpha + \omega_e i_\beta)sin2\theta_e \end{bmatrix} \\
\Phi_{\alpha2} &= \begin{bmatrix} \frac{1}{2}\rho i_\alpha - (\frac{1}{2}\rho i_\beta - \omega_e i_\alpha)sin2\theta_e - (\frac{1}{2}\rho i_\alpha + \omega_e i_\beta)sin2\theta_e \\ -\omega_e sin\theta_e \end{bmatrix} \\
\Phi_{\beta1} &= \begin{bmatrix} \frac{1}{2}\rho i_\beta + (\frac{1}{2}\rho i_\beta - \omega_e i_\alpha)cos2\theta_e - (\frac{1}{2}\rho i_\alpha + \omega_e i_\beta)sin2\theta_e \\ \omega_e cos\theta_e \end{bmatrix} \\
\Phi_{\beta2} &= \begin{bmatrix} i_\beta \\ \frac{1}{2}\rho i_\beta - (\frac{1}{2}\rho i_\beta - \omega_e i_\alpha)cos2\theta_e + (\frac{1}{2}\rho i_\alpha + \omega_e i_\beta)sin2\theta_e \end{bmatrix}
\end{aligned}
\tag{7}
$$

Due to the under-ranking problem, the approach in this paper is to identify L_q and ψ_f as fixed values of R_s and L_d, or L_q and ψ_f as fixed values of R_s and L_d.

The whole implementation process of online parameter identification is shown in Figure 1. Here is an example of identifying resistance and d-axis inductance. First, $\Phi_{\alpha1}$ and $\Phi_{\alpha2}$ are calculated from the collected i_α、i_β、ω_e、θ_e. The $\Phi_{\alpha1}$ is fed into the RLS algorithm as the input to the system and $u_\alpha - \theta_{\alpha2}\Phi_{\alpha2}$ is calculated from the reference voltage u_α and fed into the RLS algorithm as the output of the system. The discriminative values of R_s and L_d can be calculated by the equation of the least squares described in Equation (3).

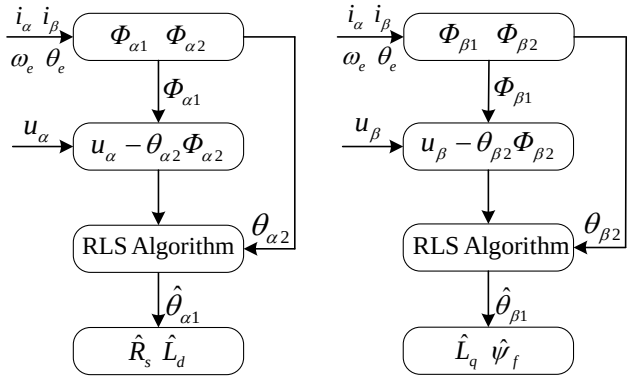

Figure 1. Implementation block diagram of online parameter identification.

3. Simulation and Experimental Results

In order to verify the correctness of the online parameter identification algorithm proposed in this paper, a permanent magnet motor model was built using Simulink in Matlab for simulation verification, and a 2 kW permanent magnet synchronous motor pair-towing experimental platform was built on which the experimental verification of the strategy proposed in this paper was carried out. The motor parameters used in the simulation and experiment are shown in Tables 1 and 2. The switching frequency is 2 kHz and the control chip of the hardware platform is TMS320F28335 from TI.

Table 1. Motor parameters used in the simulation.

Parameters	*Value*	Parameters	*Value*
Rated power	250 kW	Rotor magnetic chain	0.71 Wb
Rated frequency	126.6 Hz	Stator resistance	0.038 Ω
Rated voltage	936 V	d-axis inductor	1.67 mH
Rated current	179 A	q-axis inductor	4.02 mH

Table 2. Motor parameters used in the experiment.

Parameters	*Value*	Parameters	*Value*
Rated power	2 kW	Rotor magnetic chain	0.5962 Wb
Rated frequency	66.7 Hz	Stator resistance	1.55 Ω
Rated voltage	306 V	d-axis inductor	13.77 mH
Rated current	4.1 A	q-axis inductor	51.88 mH

3.1. Simulation results and analysis

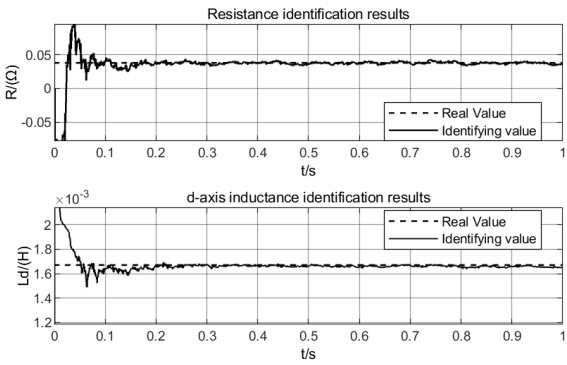

Figure 2. Resistance and d-axis inductance results.

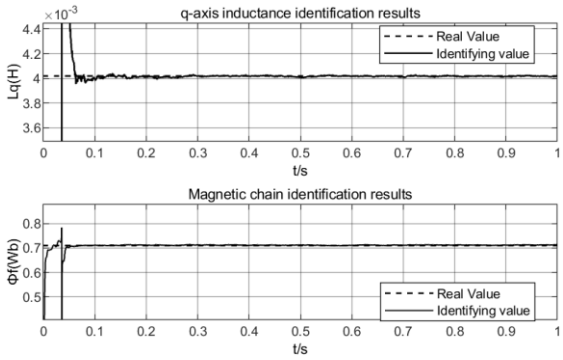

Figure 3. d-axis inductance and magnetic chain results.

Figure 2 shows the analog identification of the stator line impedance and the d-axis inductor, while Figure 3 shows the analog identification of the q-axis inductor and the magnetic field. In the case of 0.03761, the resistance of the stator is in the range of -1.1%. Under the deviation of -1.1%, the d-axis induction coefficient is 0.001651. The obtained q-axis resistance is 0.004018, and the measured result has a negative deviation of 0.05% from the measured value. The magnetic chain identification value is 0.7135, with an error of 0.5%. The convergence time is between 0.1 s and 0.2 s. Under normal conditions, the algorithm converges quickly and the error of the results is very small, so it can effectively identify the parameters.

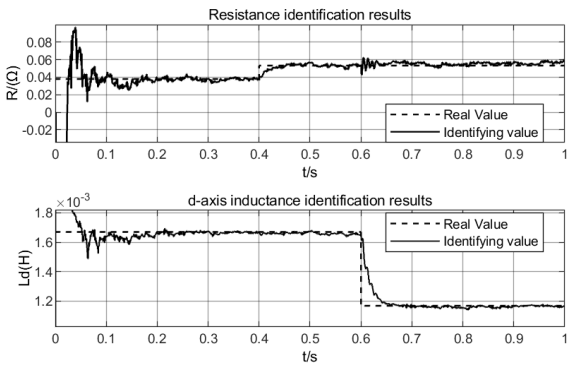

Figure 4. Results of resistance and d-axis inductance during sudden parameter changes.

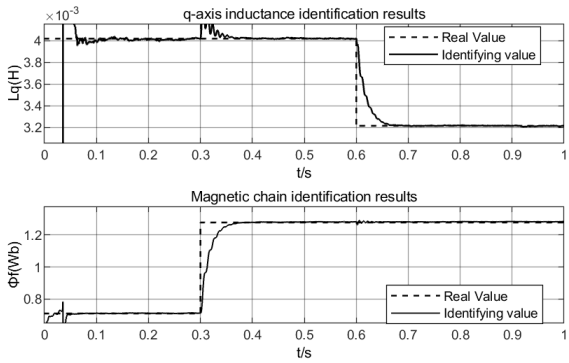

Figure 5. Results of q-axis inductance and magnetic chain during sudden parameter changes.

Secondly, the following performance of the least squares method is verified. Considering that the parameters will change in the actual working conditions, the parameters are changed in size during the motor operation in the simulation. In Figure 4, the resistance is increased by 40% at 0.4 s and the d-axis inductance is decreased by 30% at 0.6 s. In Figure 5, the q-axis inductance is reduced by 20% at 0.6 s and the magnetic chain is increased by 80% at 0.3 s. From the figure, it can be seen that the discrimination can still follow and converge quickly after the parameter change.

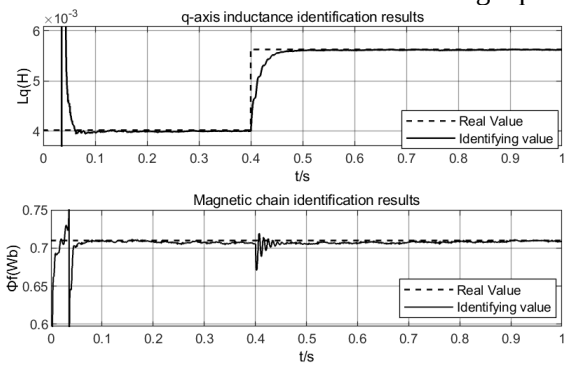

Figure 6. Identification results of q-axis inductance and magnetic chain when fixed parameters are not allowed.

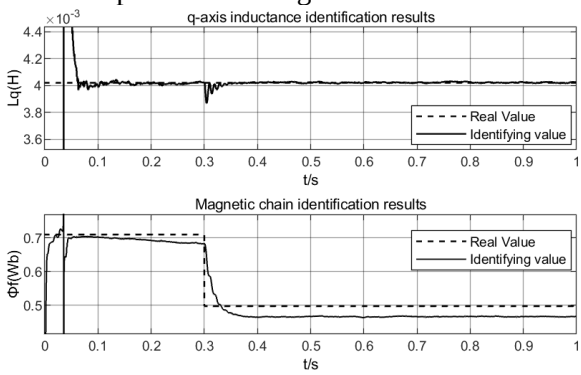

Figure 7. Identification results of q-axis inductance and magnetic chain when fixed parameters are not allowed.

Because the approach of this paper is to fix two parameters to identify the other two parameters, and the fixed two parameters, for example, are provided by the manufacturer or measured offline, and the real parameters often have some deviation, the initial value is slightly adjusted in the simulation. In Figure 6, the initial value of the resistance is changed to 1.2 times, and in Figure 7, the initial value of the d-axis inductance is changed to 0.9 times. As can be seen from the figures, although the recognition accuracy has a little deviation compared with the ideal situation, the overall stability is maintained and the error is within the acceptable range.

3.2. Experimental results and analysis

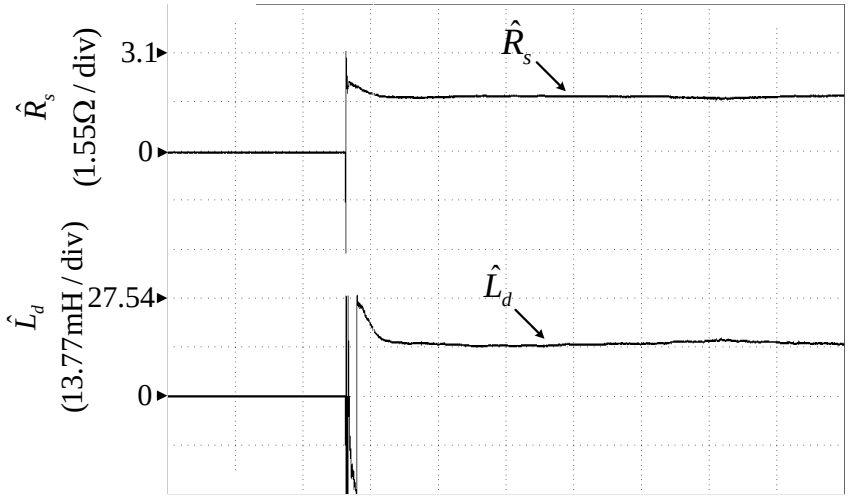

Figure 8. Identification curve of resistance and d-axis inductance when the motor runs at 30 Hz.

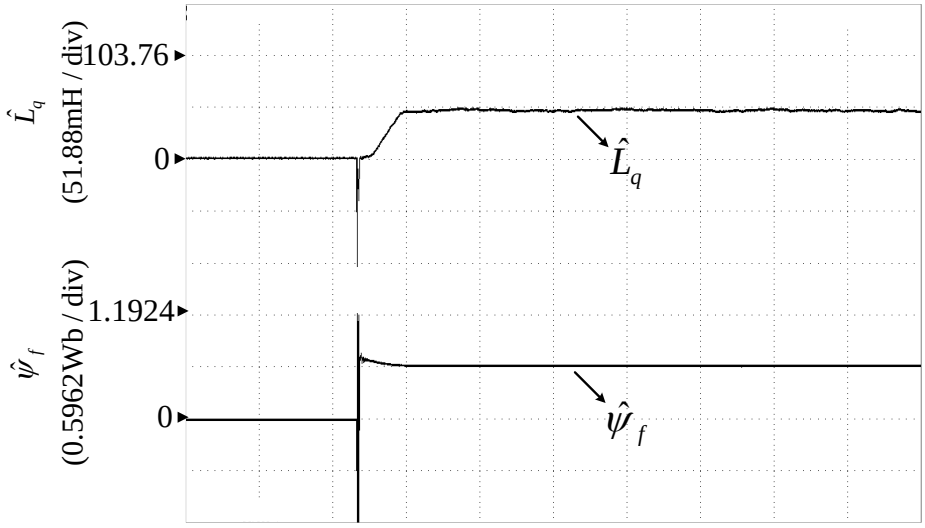

Figure 9. Identification curve of q-axis inductance and magnetic chain when the motor runs at 35 Hz.

Figure 8 shows the identification results of resistance and d-axis inductance when the motor is running at 30 Hz, and Figure 9 shows the identification results of q-axis inductance and magnetic chain when the motor is running at 35 Hz. As can be seen from the chart, using the least square method to identify the parameters of the PMSM, in the initial stage, the deviation value is relatively large, but it can quickly approach the actual value and has a very low steady-state deviation, which shows that this recognition method has high recognition accuracy and fast kinetic response-ability.

4. Conclusions

A new algorithm for dynamic characteristic identification of the PMSM is introduced. Firstly, the mathematical model of Alberta-Beta steady state in the PMSM system is established, and the parameters of the PMSM system are identified by the least square method. In view of the shortcomings of this method, some parameters are controlled first, and then other parameters are identified. The experimental results show that the proposed algorithm has the advantages of fast convergence, a small amount of calculation, and high accuracy.

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